

Ashwin Chembu

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PROFILE

Backend and full-stack engineer with 2+ years of full-time experience building Python services, AWS integrations, healthcare data workflows, reliable APIs, and tested cloud products.

TECHNICAL SKILLS

Languages & Services: Python, TypeScript, JavaScript, SQL, FastAPI, Node.js, Express, REST APIs

Cloud & Data: AWS S3, Snowflake, MongoDB, MySQL, Redis, document ingestion, data-quality workflows

Full Stack & Quality: React, pytest, Vitest, Testing Library, Playwright, UAT, workflow validation

EXPERIENCE

Business Technology Solutions Associate | ZS Associates

South San Francisco, CA | Jun 2024-Present

Healthcare AI, data, API integration, and workflow delivery

Python | FastAPI | React | Snowflake | AWS | pytest

- Built Snowflake SQL and Python service workflows across **327K+ healthcare records**, joining identifiers, specialties, and addresses with API connectivity and producing auditable exception reports for regulated product and stakeholder decisions.
- Designed reconciliation services across **67K+ healthcare records** using NPI, DEA, HIN, 340B, licenses, addresses, and fuzzy matching, routing ambiguous entities into deterministic stewardship review instead of silently publishing uncertain matches.
- Defined a **5-layer healthcare data architecture** from landing through publication, adding data-quality rules, delta processing, history tracking, and source-to-target mappings so service outputs remained maintainable, traceable, audit-ready, and understandable to technical and product stakeholders.
- Implemented Python/Flask services across **3 cloud workflow components** - API endpoints, AWS S3 storage, and Smartsheet integrations - automating document handoffs while preserving requirements, UAT evidence, and implementation documentation.
- Validated **2 enterprise workflow layers** - live CRM behavior and application metadata - then converted discrepancies into acceptance criteria, prioritized backlog items, test evidence, and implementation-ready artifacts for product owners, engineers, and internal stakeholders.

Data Analyst / Web Development Lead | Tech4Good @ UCSC

Mar 2023-Present

Collaborative research and classroom workflow tooling

TypeScript | Node.js | Express | MongoDB | Python | Figma APIs

- Contributed **158 GitHub commits**, including **44 backend-scoped commits** across TypeScript and Figma API chat, polling, Likert, text-entry, and export workflows in a collaborator-owned research toolkit, preserving accurate team ownership, public attribution, and research delivery evidence.
- Implemented **20+ Git-attributed Node/Express REST routes** backed by three MongoDB databases, populated message logs, change streams, and persisted poll and voter records, reliably synchronizing threads, edits, anonymous responses, votes, grouped responses, reveal operations, and live updates.
- Analyzed **43,340 anonymized interaction rows** across **2,805 session IDs** with Python and Pandas notebooks, while six team-published Figma Community resources reached approximately **12.2K aggregate uses** with team-library attribution preserved for reproducible analytics and public research reporting.

Founder and Lead Developer | Tallyrus

Jun 2023-Present

Full-stack retrieval tooling for educator feedback

Python | React | TypeScript | Flask | MongoDB | LangChain

- Built a retrieval-augmented grading workflow across **4 stages** - document ingestion, Pinecone vector search, rubric scoring, and educator feedback - connecting **Python NLP services** to **React interfaces** that preserved consistent scoring logic, actionable review loops, and grading decisions for educators.
- Implemented **5 application safeguards** across the React/TypeScript and Express product, including Google OAuth, protected routes, secure sessions, rate limiting, and Helmet security headers, while MongoDB and S3 managed application and document data for retrieval-based grading workflows.
- Maintained **3 backend data paths** - MongoDB application state, S3 document storage, and Pinecone retrieval - through Python and Flask services, keeping ingestion, rubric scoring, and educator feedback synchronized across tested document workflows.

PROJECTS

Private Filedrop | AI-Integrated File Exchange

2026

Secure APIs and cross-session AI coordination

Node.js | Express | AWS S3 | OpenAPI | MCP | HMAC

- Exposed **4 secure backend interfaces** - REST, OpenAPI, MCP, and signed webhooks - over local and S3-compatible storage, giving browser, API, and AI clients documented upload, versioning, messaging, coordination, and durable migration paths across development and cloud environments.
- Verified **2 webhook security checks** using HMAC-SHA256 signatures and timing-safe comparison, then documented local and cloud deployment paths so browser, API, and automation integrations could be tested repeatably without exposing storage credentials or weakening audit boundaries.
- Supported **2 storage drivers** - local and S3-compatible - with documented migration tooling, upload validation, version history, and unguessable download links for durable service operation across development and cloud environments, repeatable recovery, and controlled file handoffs.
- Documented **4 service operating modes** - local use, Render cloud deployment, ChatGPT Actions, and MCP clients - with environment-based setup and migration guidance for repeatable storage, messaging, testing, and cross-session coordination.

Forge | Personal Fitness Intelligence Backend

2026

Health-data integrations, scheduled services, and adherence analytics

Python | FastAPI | MongoDB Atlas | Render | REST APIs | webhooks

- Integrated **4 health-data sources** - Hevy, Oura, MyFitnessPal-derived Apple Health data, and Health Auto Export webhooks - through Python and FastAPI services into MongoDB Atlas for unified training, nutrition, sleep, heart-rate, recovery, and adherence analysis.
- Implemented a **6-hour synchronization cycle** plus on-demand refresh and CSV ingestion, giving the backend repeatable scheduled and user-triggered paths for updating external health records without manual data consolidation, stale dashboards, or duplicated ingestion steps.
- Deployed **1 FastAPI service** on Render with MongoDB Atlas and environment-based integration setup, then compared observed performance with a **12-week hypertrophy plan** to score workout, nutrition, and recovery adherence through documented cloud service paths.

Sylk | Multi-Tenant Supply-Chain Platform

2025-Present

Role-aware web and Android product with tested tenant isolation

React 19 | TypeScript | Firebase | Capacitor 7 | Playwright

- Built **6 core multi-tenant workflows** covering organization onboarding, shipments, fulfillment, tracking, file collaboration, and real-time chat, separating retailer-only financial data from shared manufacturing views while supporting retailer administrators, factory viewers, and staff across mobile interfaces.
- Enforced permissions for **3 role groups** through Firestore and Storage security rules, separating shared manufacturing workflows from private financial data, then validated cross-tenant access, security rules, and complete end-to-end user journeys with Vitest, **Testing Library**, Firebase emulators, and Playwright.
- Implemented **2 shipment controls** through Firebase Cloud Functions - calculation logic and validated fulfillment transitions - while React 19 and TypeScript kept onboarding, inventory, files, chat, and role-aware operational views connected across retailer and factory workflows.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7

Davis, CA | Graduated Jun 2025

Relevant Foundation: Algorithms, Databases, Distributed Systems, Operating Systems, AI/ML, Data Structures

PUBLICATION

ICER 2024 - Evaluating Exploratory Reading Groups for Supporting Undergraduate Research Pipelines in Computing - Research examining how structured exploratory reading groups support undergraduate computing research pipelines.

LINKS

Portfolio: <https://ashchembu.com> | LinkedIn: [linkedin.com/in/ashwin-chembu-a5292b205](https://www.linkedin.com/in/ashwin-chembu-a5292b205) | GitHub: github.com/ashwinchembu